

Smart Farming AI Agent

Agriculture Knowledge Base for RAG Demonstration

Purpose: This document provides general agricultural knowledge for an AI assistant demonstration. It is reference material, not a substitute for advice from local agricultural officers, agronomists, soil laboratories, or other qualified experts.

1. Crop Selection and Seasonal Planning

- Crop choice should consider season, rainfall, temperature, soil type, irrigation availability, local market demand, and the farmer's available resources.
- For kharif conditions in much of India, commonly grown crops include rice, maize, cotton, soybean, groundnut, millets, pulses and vegetables depending on the region.
- For rabi conditions, common crops include wheat, chickpea, mustard, lentil and selected vegetables depending on climate and irrigation.
- In Karnataka, crop suitability varies substantially by district. A recommendation should therefore request the farmer's district/taluk, soil type, irrigation availability and intended season before giving a specific recommendation.

2. Soil and Nutrient Management

- Soil testing is an important first step before making fertilizer recommendations. A soil test can provide information such as pH and available nutrients.
- Organic matter can be improved through practices such as compost, well-decomposed farmyard manure, crop residues and suitable green-manure crops.
- Fertilizer application should be based on crop requirement and soil-test results rather than applying the same dose to every field.
- Very acidic or alkaline soils can restrict nutrient availability. Soil amendments should be selected according to soil-test results and local recommendations.

3. Water and Irrigation Management

- Efficient irrigation aims to provide the crop with adequate water while reducing runoff, evaporation and waterlogging.
- Drip irrigation can be useful for many row crops, horticultural crops and vegetables because water can be delivered near the root zone.
- Sprinkler irrigation can be useful for several crops and soils where suitable.
- Irrigation timing should consider crop growth stage, soil moisture, weather and rainfall forecasts. Avoid unnecessary irrigation immediately before substantial rainfall when practical.

4. Integrated Pest Management (IPM)

- Integrated Pest Management combines monitoring, prevention, cultural practices, mechanical methods, biological control and, when necessary, carefully selected pesticides.

- Farmers should regularly inspect leaves, stems, flowers and fruits for insects, eggs, larvae, disease symptoms and abnormal growth.
- Examples of preventive practices include crop rotation, field sanitation, resistant or tolerant varieties where available, balanced fertilization and removal of severely affected plant material when recommended.
- A pest or disease should not be diagnosed from one symptom alone. The assistant should ask for crop age, affected plant part, symptoms, location and photographs when available.

5. Common Crop Problems and Safe Guidance

- Yellowing leaves can have several causes, including nutrient deficiency, water stress, root problems, pests or disease. A diagnosis requires additional field information.
- Wilting can result from insufficient water, root damage, vascular disease, waterlogging or other stresses. Check soil moisture and inspect roots and stems before recommending treatment.
- Leaf spots, powdery growth, curling, holes and discoloration may be associated with different pests or diseases. Avoid recommending a specific pesticide solely from a vague symptom.
- When pesticide use is considered, the farmer should follow the product label, approved crop/pest use, dosage, safety interval and protective-equipment requirements, and seek local agricultural guidance when diagnosis is uncertain.

6. Weather-Aware Farming

- Weather information can support decisions about sowing, irrigation, spraying and harvesting.
- Rainfall probability, temperature, humidity, wind and extreme-weather alerts can affect crop management.
- An AI assistant must not invent current weather conditions. If a live weather service is not connected, it should clearly state that the answer is based on general knowledge and ask the farmer to provide current weather information.

7. Market and Mandi Information

- Crop prices can vary by market, date, quality, variety, arrival volume and demand.
- An AI assistant must not invent a current mandi price. If live market data is unavailable, it should clearly say that real-time prices cannot be verified.
- For a price-based recommendation, useful information includes crop/variety, quality or grade, quantity, location and target market.

8. Localized Farmer Interaction

- The assistant should communicate in the farmer's preferred language and use simple, practical explanations.
- For a localized recommendation, collect relevant information such as state, district/taluk, village if appropriate, crop, crop age, soil type, irrigation source, recent rainfall, symptoms and farming objective.
- The assistant should distinguish between general guidance and recommendations that require field inspection or expert confirmation.

9. RAG Response Rules for the Smart Farming Agent

- The assistant should use retrieved knowledge from this document when it is relevant to the farmer's question.
- It should not fabricate facts that are absent from the retrieved context.
- For current weather, current market prices or other live information, it should state when live data is unavailable rather than guessing.
- It should provide practical steps, explain uncertainty, and recommend consultation with a local agriculture expert for high-risk or uncertain cases.
- When useful, the response should follow this structure: **Assessment** → **Recommended actions** → **What to monitor** → **When to seek expert help**.

10. Example Farmer Questions

- Which crops are suitable for my soil and the coming season?
- My tomato leaves are curling and turning yellow. What information should I check first?
- How can I reduce water use in vegetable cultivation?
- What is integrated pest management and how can I apply it?
- What information do you need before recommending fertilizer?
- What information is required to give a reliable crop recommendation for a farmer in Karnataka?
- What should I provide if I want help identifying a crop pest from a photograph?